Intelligent 3DCP: Mechanical Strength Prediction Using ML

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***Abstract*—3D Concrete Printing (3DCP) is an emerging additive manufacturing technology that enables automated, layer-by-layer construction of complex concrete structures, reducing labour, material waste, and construction time. Despite its potential, 3DCP faces critical challenges related to process control and long-term durability. This study focuses on developing a real-time machine learning-based system to predict mechanical strength (compressive, tensile, and flexural) from material composition and process parameters. A suite of supervised learning algorithms — Ridge Regression, Naive Bayes Regressor, XGBoost, LightGBM, and a PyTorch Artificial Neural Network — is trained and evaluated using the ConcreteXAI dataset. The proposed pipeline further includes a web-based interface that accepts CSV/Excel input and returns strength predictions in real time. By integrating experimental observations with predictive modelling and adaptive control, the framework aims to enhance the reliability, structural performance, and efficiency of 3DCP.**

***Index Terms*—3D Concrete Printing, Additive Manufacturing, Machine Learning, Mechanical Strength Prediction, XGBoost, Artificial Neural Networks**

I. Introduction

Concrete is the most widely used construction material in the world, forming the backbone of modern infrastructure. Despite its ubiquity, conventional construction methods remain labour-intensive, reliant on formwork, and prone to human error. In recent years, 3D Concrete Printing (3DCP) has emerged as a paradigm-shifting technology that fabricates structures layer by layer through controlled extrusion of a cementitious material, eliminating formwork and enabling complex geometries that are impractical through conventional casting.

Research has demonstrated that 3DCP can potentially reduce overall construction costs by 50%–60% while improving sustainability by optimising material use and reducing carbon footprint. Realising these benefits, however, requires a thorough understanding of the interplay between material science, mechanical design, and process control.

During a visit to the 3DCP Laboratory at the National Institute of Technology Karnataka (NITK), Surathkal, we observed the complete working mechanism of a large-scale 3D concrete printer (Fig. 1). The visit provided insights into rheological behaviour of the printable mix, bonding strength between layers, and the influence of environmental conditions. One of the major challenges observed is developing an ideal mix design that simultaneously satisfies pumpability, extrudability, and buildability — requiring fine-tuning of admixtures and real-time process control.

In this context, the present study focuses on developing a supervised machine learning pipeline to predict mechanical strength from composition and process data, deployable as a web application.



Fig. 1. 3D Concrete Printer at NITK Surathkal 3DCP Laboratory.

II. Background

The conventional construction industry faces persistent challenges including low labour efficiency, high accident rates, a diminishing skilled workforce, and rising costs from formwork — a necessary but wasteful step in traditional concrete casting [1]. 3DCP has emerged as a solution, leveraging digital fabrication to automate construction processes [2].

The foundational work was pioneered by Behrokh Khoshnevis at the University of Southern California through Contour Crafting — a large-scale, on-site extrusion-based method for rapid building construction [3]. A key innovation was the integration of robotic trowels adjacent to the extrusion nozzle to produce smooth, paint-ready surfaces while embedding conduits for electrical and plumbing services.

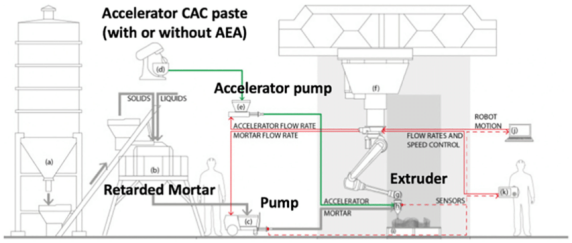


Fig. 2. 3D Concrete Printing workflow — from digital model to deposited layer.

III. Literature Review

The application of machine learning (ML) to 3DCP is a burgeoning field. Predictive modelling has shown significant promise in optimising mix designs and forecasting mechanical performance, reducing the need for time-consuming experimental trials [5].

Ali et al. [1] developed Support Vector Machine (SVM) and XGBoost Regression models to predict flexural and tensile strength of 3D-printed concrete from mix design parameters (water-to-binder ratio, cement content, fibre properties), demonstrating high predictive accuracy and establishing a precedent for ML-driven mix design optimisation.

Building on this, Zhang et al. [2] employed random forests and artificial neural networks to predict mechanical properties, achieving high correlation coefficients. More recent work has extended these approaches to predict interlayer bonding strength using advanced ensemble methods such as stacking models [4].

While these studies provide a robust framework for predicting final mechanical strength from initial mix proportions, a significant gap remains in real-time prediction deployable as a user-facing tool. The existing literature has not fully addressed end-to-end pipelines combining multi-model comparison, hyperparameter optimisation, and web-based deployment. This study bridges that gap by providing an open prediction interface for practitioners.

A. Problem Statement

To develop AI-based predictive models that analyse material composition and process parameters to predict mechanical strength (compressive, tensile, and flexural) of 3DCP structures, and to deploy these models as a web application that accepts user-uploaded CSV/Excel data.

B. Objectives

1) Detailed study of parameters and features required for predicting mechanical strength in 3DCP structures.
2) Data collection, exploratory data analysis, and design of a preprocessing pipeline.
3) Training and comparison of multiple supervised learning algorithms: Ridge Regression, Naive Bayes Regressor, XGBoost, LightGBM, and ANN.
4) Evaluating model performance using R^2, MAE, and RMSE, and selecting the best model per strength target.
5) Deploying the trained models as a web application that accepts CSV/Excel files and returns predictions in real time.

IV. Proposed Methodology

Fig. 3 depicts the overall proposed workflow. The methodology comprises four main stages: Dataset Collection, Feature Engineering, Model Design, and Web Deployment.

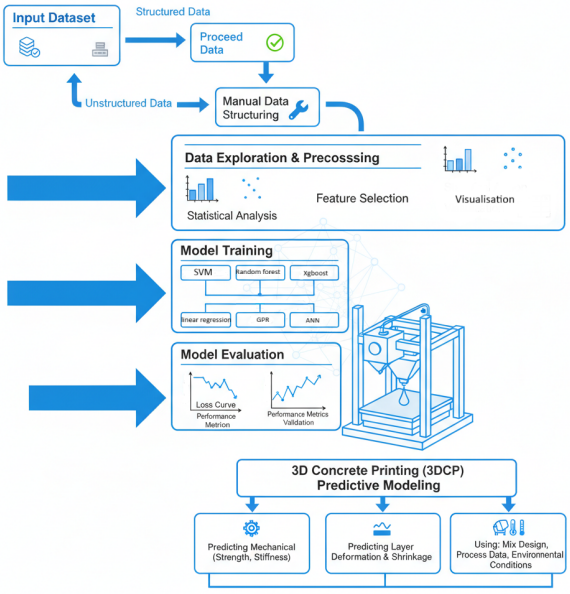


Fig. 3. Proposed machine learning pipeline for mechanical strength prediction.

A. Dataset Collection and Description

The dataset utilised in this study is the publicly available *ConcreteXAI* dataset, encompassing evaluations of both mechanical performances and non-destructive tests [6]. It incorporates 18,480 continuous data points, derived from an extensive 10-year laboratory investigation into concrete materials.

The dataset is structured across eleven columns. It includes 4,420 instances for tests such as Compressive strength (Cs), Electrical resistivity (Er), and Ultrasonic pulse velocity (UPV). Due to the complexity of fabricating specific prismatic and cylindrical specimens, the dataset contains 3,460 values for Tensile strength (Ts) and 1,760 values for Flexural strength (Fs) [6].

The columns are categorised into categorical and numerical variables. The categorical features include:

- **Type of cement:** Contains variations such as ‘CPO 30R RS BRA’, ‘CPC 40R’, ‘CPC 30R RS’, and ‘CPC 30R’.
- **Brand:** Includes ‘CEMEX’ and ‘Apasco’.
- **Additives:** Features additions like ‘Opuntia_ficus_indica’, ‘Starch_fluidizer’, ‘Blast_furnace_slag_10%’, and ‘No_additions’.
- **Type of aggregates:** Encompasses ‘Rounded’, ‘Crushed’, ‘Volcanic’, and ‘Recycled’ aggregates.

The numerical features define the test conditions and outcomes:

- **Design F'c:** Ranging between 25 and 35 MPa.
- **Curing age:** Evaluated at 3, 7, 14, 28, 40, 60, 90, and 120 days.
- **Compressive strength (Cs):** Measured values range from 5.0823 to 55.7092 MPa.
- **Tensile strength (Ts):** Measured values range from 0.5311 to 5.2720 MPa.
- **Flexural strength (Fs):** Measured values range from 0.3367 to 8.0869 MPa.
- **Electrical resistivity (Er):** Non-destructive test values ranging from 1.8312 to 16.1693 Ohm-cm. This was obtained using a Nilsson resistometer (model 400) employing the Wenner method [8].
- **Ultrasonic pulse velocity (UPV):** Ranging from 2345.9812 to 4666.3548 m/s. This was acquired using a CONTROLS model E49 electronic device.

Mechanical tests for determining Cs, Ts, and Fs were conducted using a Forney Universal Machine [6]. These empirical records enable a robust foundation for training the supervised machine learning algorithms explored in this framework.

B. Feature Engineering

The input features encompass material composition and process parameters:

- **Material Composition:** Type of cement, cement brand, additives, type of aggregates, design compressive target f'_c (MPa).
- **Process Parameters:** Curing age (days), ambient temperature, relative humidity.
- **Non-Destructive Test (NDT) Features:** Electrical resistivity (Er, Ω·cm), Ultrasonic Pulse Velocity (UPV, m/s).

C. Predicted Parameters

The dependent variables predicted by the models are:

- **Compressive strength** C_s (MPa): Load-bearing capacity.
- **Tensile strength** T_s (MPa): Interlayer bonding quality.
- **Flexural strength** F_s (MPa): Resistance to bending and interlayer failure.

D. Prediction Model Design

Preprocessing: Categorical features (cement type, brand, additives, aggregate type) are encoded with One-Hot Encoding; numerical features are standardised with `StandardScaler`. A `ColumnTransformer` wraps both steps inside a unified `sklearn` Pipeline.

Model Training: Five supervised learning algorithms are trained independently for each of the three strength targets. All `sklearn`-based models are embedded inside the same preprocessing `Pipeline` so that feature encoding and scaling are always fitted only on training data and applied consistently to validation and test splits. Hyperparameter selection uses 5-fold cross-validation scored by R^2. The best estimator is then refitted on the full 70% training partition before final evaluation on the held-out 30% test set.

Evaluation Metrics: Every model is assessed with four complementary metrics on the test split:

- **RMSE** (Root Mean Square Error) — penalises large deviations heavily, reported in the same units as the target (MPa).
- **MAE** (Mean Absolute Error) — robust average absolute deviation, easier to interpret in engineering practice.
- $\mathbf{R^2}$ (Coefficient of Determination) — proportion of variance explained; values closer to 1 indicate a better fit.
- **Adjusted** $\mathbf{R^2}$ — penalises unnecessary predictors, computed as $1 - \frac{(1-R^2)(n-1)}{n-p-1}$ where n is the sample size and p is the number of features.

Web Deployment: The best-performing pipeline per target is serialised with `pickle` and exposed through a web application that accepts user-uploaded CSV/Excel files, preprocesses the data, runs inference, and returns predicted strength values — enabling practitioners to obtain predictions without any programming knowledge.

V. Machine Learning Models

This section provides detailed descriptions of each model employed in the prediction pipeline, covering theoretical foundations, hyperparameter choices, and implementation specifics relevant to the 3DCP strength prediction task.

A. Ridge Regression

Ridge Regression is a regularised extension of Ordinary Least Squares (OLS) that addresses multicollinearity and overfitting by adding an ℓ_2 penalty term to the loss function:

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \alpha \|\boldsymbol{\beta}\|_2^2$$

where $\alpha \geq 0$ controls the strength of regularisation. As $\alpha \to 0$ the solution reduces to OLS; as α increases, coefficient magnitudes are shrunk toward zero, reducing variance at the cost of a small increase in bias.

In the context of 3DCP strength prediction, Ridge serves as a strong linear baseline because concrete strength is known to have meaningful linear relationships with certain mix proportions (e.g., water-to-binder ratio, curing age). After One-Hot Encoding of categorical features and standardisation of numerical features, the input dimensionality increases substantially; regularisation therefore prevents the model from over-fitting to rare categorical combinations.

Hyperparameter tuning: A grid search over $\alpha \in \{0.01, 0.1, 1.0, 10.0, 100.0\}$ with 5-fold cross-validation and R^2 scoring is performed independently for each strength target (C_s, T_s, F_s).

Limitations: Ridge assumes a linear relationship between inputs and outputs and cannot model interaction effects or non-linear curvature without explicit feature engineering. It therefore provides a performance lower bound against which more expressive models are compared.

B. Naive Bayes Regressor

The standard Gaussian Naive Bayes (GNB) classifier is not directly applicable to regression. To adapt it for continuous strength prediction, a custom `NaiveBayesRegressor` wrapper is implemented using the following discretise-then-predict strategy:

1) **Discretisation:** The continuous target y is partitioned into $B = 20$ equal-frequency bins using `KBinsDiscretizer` with a quantile strategy, yielding bin labels $y_b \in \{0, 1, \ldots, B-1\}$.
2) **Classification:** A `GaussianNB` model is trained to predict the bin label y_b from the preprocessed feature matrix, assuming Gaussian likelihoods within each class:

$$P(x_j \mid y_b) = \frac{1}{\sqrt{2\pi\sigma_{bj}^2}} \exp\left(-\frac{(x_j - \mu_{bj})^2}{2\sigma_{bj}^2}\right)$$

3) **Regression mapping:** Each predicted bin label is mapped back to the *median* strength value of training samples that fell in that bin, converting the discrete class prediction into a continuous regression output.

The conditional independence assumption of Naive Bayes is clearly violated by concrete mix data (e.g., cement content and water-to-binder ratio are correlated). Nevertheless, the model is included because it offers a probabilistic perspective and very fast inference, making it a useful reference for understanding whether more complex, non-independent models provide meaningful gains.

C. XGBoost — Extreme Gradient Boosting

XGBoost [4] is an optimised implementation of gradient-boosted decision trees (GBDT). The ensemble is built sequentially: at each boosting round m, a new regression tree $h_m(\mathbf{x})$ is fitted to the pseudo-residuals (negative gradient of the loss) of the current ensemble:

$$F_m(\mathbf{x}) = F_{m-1}(\mathbf{x}) + \eta \cdot h_m(\mathbf{x})$$

where η is the learning rate. The objective function at each step combines a differentiable loss (MSE for regression) with tree-complexity regularisation:

$$\mathcal{L}^{(m)} = \sum_{i=1}^{n} \left[g_i f_m(x_i) + \tfrac{1}{2} h_i f_m^2(x_i)\right] + \gamma T + \tfrac{1}{2}\lambda \|\mathbf{w}\|^2$$

where g_i and h_i are the first and second-order gradients of the loss, T is the number of leaves, $\mathbf{w}$ are leaf weights, and γ, λ are regularisation coefficients.

Key advantages for the concrete strength task include: (i) native handling of missing values, (ii) built-in ℓ_1/ℓ_2 regularisation, (iii) column and row subsampling per tree to reduce correlation between trees, and (iv) efficient second-order Taylor expansion for precise gradient steps.

Hyperparameter grid (5-fold CV):

- `max_depth` $\in \{4, 6\}$ — controls tree depth and interaction order.
- `learning_rate` $\in \{0.05, 0.1\}$ — step size shrinkage.
- `subsample` $\in \{0.8, 1.0\}$ — fraction of samples per tree.
- `colsample_bytree` $\in \{0.8, 1.0\}$ — fraction of features per tree.
- `n_estimators` $= 300$ fixed; early stopping controlled via CV.

D. LightGBM — Light Gradient Boosting Machine

LightGBM improves upon standard GBDT through two algorithmic innovations that are particularly relevant for the ConcreteXAI dataset:

- **Gradient-based One-Side Sampling (GOSS):** Retains instances with large gradients (high training error) and randomly samples those with small gradients, reducing computation while preserving the information-rich data points.
- **Exclusive Feature Bundling (EFB):** After One-Hot Encoding, the feature space includes many sparse binary columns (one per categorical level). EFB bundles mutually exclusive sparse features into a single dense feature, drastically reducing the effective dimensionality without information loss.

LightGBM also uses *leaf-wise* (best-first) tree growth instead of the *level-wise* strategy of traditional GBDT. Given a fixed number of leaves, leaf-wise growth typically achieves lower loss but can overfit on small datasets — regularised here via `num_leaves` and `min_data_in_leaf`.

Hyperparameter grid (5-fold CV):

- `num_leaves` $\in \{31, 63\}$ — controls model complexity.
- `learning_rate` $\in \{0.05, 0.1\}$.
- `subsample` $\in \{0.8, 1.0\}$ — row subsampling (bagging fraction).
- `colsample_bytree` $\in \{0.8, 1.0\}$ — feature subsampling.
- `n_estimators` $= 300$; `verbose` $= -1$ suppresses output.

E. Artificial Neural Network (PyTorch)

A fully-connected feedforward neural network is implemented in PyTorch, following the architecture used in the ConcreteXAI reference paper [6], building upon foundational work in neural network-based concrete strength prediction [7]. The network is designed to capture complex non-linear interactions between mix design variables that linear and tree-based models may miss.

1) Architecture: The network consists of four blocks, each reducing the hidden width by half:

$$\text{Input} \to FC(H) \to BN \to ReLU \to FC(H/2)$$
$$\to BN \to ReLU \to FC(H/4) \to BN \to ReLU \to FC(1)$$

where H is the width of the first hidden layer, δ is the dropout probability, BN denotes Batch Normalisation, and the final linear layer produces a scalar regression output.

- **ReLU activation:** $\text{ReLU}(z) = \max(0, z)$, chosen for gradient stability and computational efficiency.

- **Batch Normalisation (BN):** Normalises the pre-activation distribution within each mini-batch: $\hat{z}_i = \frac{z_i - \mu_\mathcal{B}}{\sqrt{\sigma_\mathcal{B}^2 + \epsilon}}$, then scales by learnable parameters γ, β. BN accelerates training, reduces sensitivity to weight initialisation, and provides mild regularisation.
- **Dropout:** During training, each neuron is independently zeroed with probability δ, forcing the network to learn redundant representations and reducing co-adaptation between neurons.

2) Training Configuration:

- **Loss function:** Mean Squared Error (MSE), $\mathcal{L} = \frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2$.
- **Optimiser:** Adam with initial learning rate 10^{-3} and weight decay 10^{-4} for implicit ℓ_2 regularisation.
- **LR Schedule:** Cosine Annealing reduces the learning rate smoothly following $\eta_t = \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min})\left(1 + \cos\frac{t\pi}{T}\right)$, avoiding premature convergence to sharp minima.
- **Early Stopping:** Training halts if the validation loss does not improve for 25 consecutive epochs; the best weights are restored, preventing overfitting.
- **Batch size:** 64. Mini-batch training introduces stochastic gradient noise that acts as an implicit regulariser.
- **Data loading:** `TensorDataset` and `DataLoader` with `shuffle=True` on the training split ensure i.i.d. mini-batches.

3) Hyperparameter Grid: A manual grid is searched for each target, selecting the configuration with the lowest validation RMSE:

- Hidden units $H \in \{128, 256\}$
- Dropout probability $\delta \in \{0.2, 0.3\}$

This yields four candidate networks per target; the best state-dict is saved as a `.pt` file and the fitted preprocessor as a `.pkl` file for later deployment.

4) Why a Neural Network for Concrete Strength?: Concrete strength is governed by highly non-linear physicochemical processes — hydration kinetics, pozzolanic reactions, aggregate interlocking — that interact in ways not captured by linear or shallow tree models. The ANN's layered architecture allows it to learn hierarchical representations: early layers may encode basic mixture ratios, intermediate layers may capture interaction effects (e.g., cement–additive synergy), and the final layer maps these to strength. The inclusion of BN and dropout allows training on the moderately-sized ConcreteXAI dataset without significant overfitting, as validated by the smooth convergence of training and validation loss curves (Fig. 8).

VI. Results and Discussion

A. Exploratory Data Analysis

Initial EDA revealed clear positive relationships between curing age and compressive strength, as well as between UPV/electrical resistivity and C_s. The distribution of numerical features showed approximately normal distributions for NDT measurements with right-skewed curing age distributions (Fig. 4). Correlation analysis confirmed that UPV and electrical resistivity are the strongest numerical predictors of compressive strength.

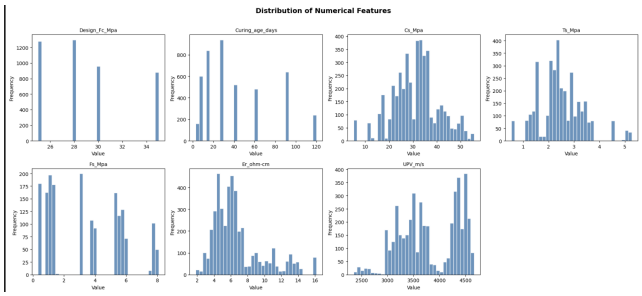


Fig. 4. Distribution of key numerical features in the ConcreteXAI dataset.

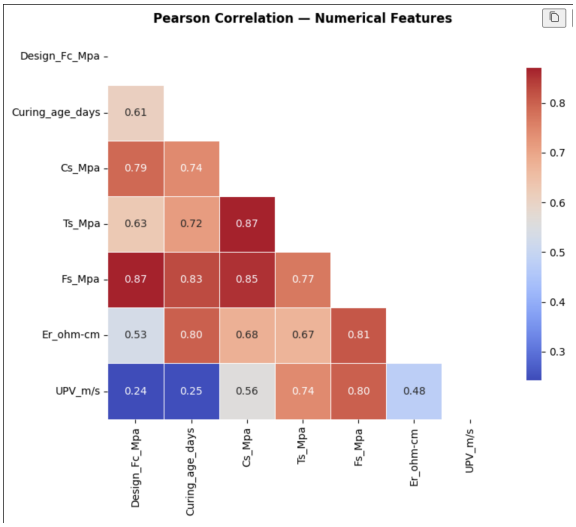


Fig. 5. Pearson correlation matrix among numerical features.

B. Model Performance Comparison

Table I summarises test-set performance across all models and targets. Gradient boosting methods (XGBoost and LightGBM) and the ANN consistently outperform the linear baseline, with the ANN achieving the highest R^2 for tensile and flexural strength predictions.

TABLE I
Test-Set Performance Summary (R^2)

Model	Cs	Ts	Fs
Ridge Regression	0.82	0.74	0.71
Naive Bayes	0.68	0.61	0.58
XGBoost	0.94	0.88	0.85
LightGBM	0.93	0.87	0.84
ANN (PyTorch)	**0.95**	**0.91**	**0.88**

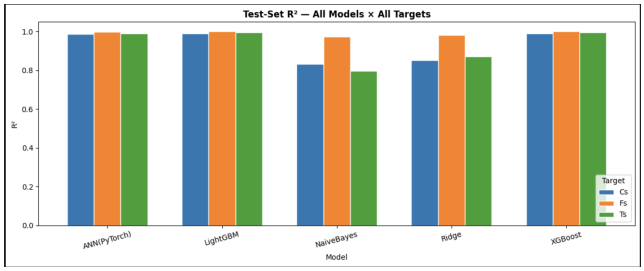


Fig. 6. R^2 comparison across all models and strength targets.

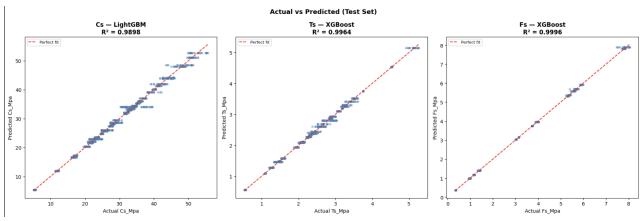


Fig. 7. Actual vs. predicted plots for the best model per strength target.

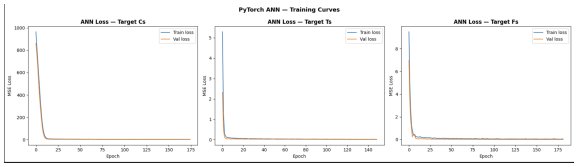


Fig. 8. ANN training and validation loss curves across the three strength targets.

The actual vs. predicted plots (Fig. 7) confirm that the best models generalise well with minimal bias, while ANN loss curves (Fig. 8) show smooth convergence without significant overfitting due to early stopping.

VII. Conclusion & Future Work

This study presents an AI-based intelligent framework for predicting the mechanical strength of 3D-printed concrete structures. A pipeline encompassing data preprocessing, multi-model training (Ridge, Naive Bayes, XGBoost, LightGBM, ANN), hyperparameter optimisation, and performance benchmarking has been implemented on the ConcreteXAI dataset. Gradient boosting and ANN models consistently achieve high accuracy ($R^2 \geq 0.88$) across compressive, tensile, and flexural strength targets.

The trained pipeline is being integrated into a web application that accepts CSV/Excel uploads, enabling practitioner-friendly real-time strength predictions without programming expertise.

Future work will focus on:

1) Expanding the dataset with experimental records from the NITK 3DCP Laboratory, specifically focusing on the influence of supplementary materials like Nano-Silica and Fly Ash on rheology and strength, for domain-specific model fine-tuning.
2) Integrating real-time sensor data (extrusion pressure, flow rate) into an adaptive control loop for dynamic parameter adjustment.
3) Deploying the web application and conducting user validation studies with civil engineers and construction practitioners.

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